

## 1. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{-4x^5 - 3x^4 - 4}{-3x^3 - 2x^2 + 1} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{-3x^2 - x + 1}{-2x^4 + 3x^3 + 1} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{x^4 + 2x^3 - 2}{-x^4 - x^3 - 2} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{x^5 - x^4 - 4}{-3x^2 + x + 2} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{-4x^5 + 2x^4 + 3}{4x^5 + x^4 - 4} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{4x^2 - 3x + 3}}{\sqrt[5]{x^4 - 3x^3 + 1}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{3x^2 + 2x - 3}}{\sqrt[4]{4x^3 + 3x^2 - 4}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{x^2 - 4x + 4}}{\sqrt[4]{4x^3 - 4x^2 - 2}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{3x^5 - 4x^4 + 4}}{\sqrt[5]{2x^3 + 4x^2 - 1}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{4x^2 + 3x - 4}}{\sqrt[4]{3x^5 + 3x^4 + 3}} =$$

11.

$$(x + 4y)^5$$

## 2. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{-x^5 + 2x^4 - 1}{2x^2 - 2x + 2} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{-3x^3 + 4x^2 + 3}{x^5 - x^4 + 4} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{-2x^4 - 2x^3 - 2}{-x^3 - 3x^2 - 2} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{3x^5 - 4x^4 - 1}{x^5 + 2x^4 - 1} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{-3x^2 - 2x + 1}{-4x^4 + 4x^3 - 3} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{x^2 + 3x - 2}}{\sqrt[4]{4x^3 + x^2 - 1}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{3x^2 + 2x + 4}}{\sqrt{x^3 + 3x^2 - 4}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{3x^3 + 3x^2 - 1}}{\sqrt[5]{2x^2 + 4x + 4}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{3x^4 - x^3 + 2}}{\sqrt[4]{3x^5 - x^4 + 4}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{3x^2 - 2x + 4}}{\sqrt[4]{3x^4 + x^3 - 3}} =$$

11.

$$(2x - 4y)^4$$

### 3. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{-x^5 + x^4 + 1}{-x^5 - 3x^4 - 3} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{x^4 + 4x^3 + 4}{-2x^5 + 4x^4 + 1} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{-2x^5 - 4x^4 + 4}{4x^4 - 2x^3 + 3} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{4x^2 - x + 2}{-4x^4 + x^3 - 2} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{-3x^4 - 3x^3 + 4}{-4x^2 - 2x + 2} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{2x^2 + 3x - 2}}{\sqrt[4]{4x^3 + x^2 + 2}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{x^4 + x^3 + 2}}{\sqrt[5]{3x^4 + 3x^3 + 2}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{x^2 + 2x + 3}}{\sqrt[4]{2x^3 + 4x^2 + 2}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{x^4 + 3x^3 + 4}}{\sqrt[5]{3x^2 - 4x - 4}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{x^2 + 3x - 2}}{\sqrt[3]{4x^2 + 3x - 3}} =$$

11.

$$(3x + 3y)^3$$

## 4. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{2x^2 - 2x + 1}{-3x^4 + 4x^3 - 3} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{-4x^2 + 3x - 2}{-3x^3 + 4x^2 + 4} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{3x^4 - 3x^3 + 2}{-x^4 - x^3 - 2} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{x^5 - 3x^4 - 2}{-4x^5 - 3x^4 - 4} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{-2x^4 - 2x^3 + 1}{-3x^4 + x^3 - 4} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{3x^2 + 2x + 2}}{\sqrt[4]{x^2 - 4x + 2}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{x^5 + x^4 + 3}}{\sqrt[3]{2x^2 + 3x + 1}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{2x^4 + x^3 - 4}}{\sqrt{x^4 + 2x^3 + 1}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{2x^2 + 3x + 1}}{\sqrt{x^5 - x^4 + 4}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{2x^3 - 3x^2 - 1}}{\sqrt{4x^3 - x^2 + 3}} =$$

11.

$$(x - 2y)^4$$

## 5. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{-3x^5 + 2x^4 - 4}{2x^2 + 3x - 1} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{-3x^2 - 3x + 1}{x^3 + x^2 + 2} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{-4x^3 + 3x^2 + 2}{-x^3 + 3x^2 - 1} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{-3x^5 + 3x^4 + 3}{-2x^4 - 3x^3 - 3} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{4x^2 + x - 2}{-3x^3 + x^2 + 4} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{x^3 + 3x^2 - 3}}{\sqrt[4]{3x^5 + 4x^4 - 4}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{x^5 - 2x^4 + 1}}{\sqrt{4x^3 - 4x^2 - 4}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{x^3 + 3x^2 - 1}}{\sqrt[3]{2x^5 + 2x^4 + 1}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{3x^5 + x^4 + 3}}{\sqrt[5]{4x^5 - 3x^4 - 2}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{4x^4 + 4x^3 + 4}}{\sqrt[4]{2x^5 + 3x^4 + 4}} =$$

11.

$$(3x + y)^4$$

## 6. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{-x^2 + 4x - 4}{3x^3 - 4x^2 - 3} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{-x^4 - 3x^3 + 1}{-x^5 + 4x^4 - 2} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{-x^4 - 4x^3 + 4}{4x^2 - 2x - 4} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{2x^4 + 3x^3 + 1}{3x^2 + 2x - 3} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{x^5 + 4x^4 + 2}{-4x^2 + 2x + 4} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{2x^2 + 2x - 4}}{\sqrt[3]{3x^2 + 4x + 4}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{2x^2 - 2x - 1}}{\sqrt[4]{2x^3 + 4x^2 + 3}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{x^5 + 2x^4 + 4}}{\sqrt[5]{3x^4 - 3x^3 - 2}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{3x^2 - 3x - 2}}{\sqrt[3]{3x^3 - 3x^2 + 2}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{3x^5 + 4x^4 - 4}}{\sqrt[5]{2x^5 - 4x^4 + 4}} =$$

11.

$$(x + y)^2$$

## 7. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{-x^2 + 2x + 1}{-4x^5 + 3x^4 - 1} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{-x^4 - 4x^3 - 1}{-4x^4 - x^3 - 3} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{-3x^5 - x^4 + 4}{-2x^5 + 4x^4 + 4} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{2x^2 - x - 4}{3x^5 - 3x^4 - 1} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{-2x^5 - x^4 - 2}{3x^3 - 2x^2 - 1} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{4x^5 - 3x^4 + 2}}{\sqrt{2x^4 + 4x^3 + 2}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{2x^3 + 2x^2 + 1}}{\sqrt[5]{x^5 - 3x^4 + 1}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{3x^3 + 4x^2 + 3}}{\sqrt[4]{4x^3 + 3x^2 - 3}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{3x^3 - x^2 - 2}}{\sqrt{4x^5 - 2x^4 - 4}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{3x^3 - 4x^2 - 4}}{\sqrt[3]{2x^4 + 3x^3 - 2}} =$$

11.

$$(4x - 4y)^4$$

## 8. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{2x^4 + 2x^3 - 2}{-x^3 - 2x^2 + 3} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{-2x^3 - 2x^2 - 4}{2x^3 - 2x^2 - 2} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{x^4 - 4x^3 + 4}{-x^3 + 3x^2 + 1} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{-2x^2 + 2x + 2}{4x^3 + 4x^2 - 1} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{-2x^4 + x^3 - 3}{4x^5 - 4x^4 + 2} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{x^4 + x^3 + 2}}{\sqrt[5]{3x^5 - 3x^4 + 2}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{2x^2 - 4x - 4}}{\sqrt[3]{4x^2 + 4x - 4}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{x^4 + 3x^3 - 3}}{\sqrt[4]{2x^3 + 4x^2 + 3}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{3x^2 - x + 3}}{\sqrt[3]{2x^3 - 3x^2 + 1}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{x^3 + 2x^2 + 2}}{\sqrt[4]{3x^2 - 3x - 4}} =$$

11.

$$(4x - y)^3$$



## 9. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{3x^3 + 3x^2 - 3}{-4x^2 - 3x + 3} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{-2x^4 - 4x^3 + 3}{-2x^4 - 4x^3 - 1} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{x^3 - 3x^2 - 3}{-3x^5 - 4x^4 + 4} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{x^2 - 2x + 4}{-2x^4 + 3x^3 - 4} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{4x^4 - 4x^3 - 2}{-3x^5 + 3x^4 + 1} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{2x^4 - x^3 - 3}}{\sqrt[5]{x^2 + 4x - 2}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{3x^3 - 3x^2 + 1}}{\sqrt[4]{3x^4 - 4x^3 + 4}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{4x^5 - x^4 + 1}}{\sqrt[4]{x^2 - 3x + 3}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{2x^4 + 4x^3 - 1}}{\sqrt[3]{2x^3 - x^2 + 4}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{2x^3 + 3x^2 - 4}}{\sqrt[4]{x^5 - 3x^4 - 3}} =$$

11.

$$(2x + 4y)^5$$

## 10. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{-x^2 - 4x - 3}{4x^4 + x^3 - 1} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{-3x^2 + 3x - 4}{-x^4 - 4x^3 - 2} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{-4x^4 - 2x^3 - 1}{-2x^5 - x^4 + 1} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{3x^2 + 3x - 4}{4x^5 + 3x^4 - 1} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{3x^2 - 2x + 4}{x^5 - 2x^4 + 1} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{x^2 - 4x + 1}}{\sqrt[5]{3x^2 - 2x + 4}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{4x^3 + 4x^2 + 2}}{\sqrt{2x^3 + 2x^2 + 4}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{3x^5 + x^4 + 2}}{\sqrt{4x^2 - x + 3}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{2x^3 + x^2 - 1}}{\sqrt{4x^2 + 2x + 1}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{4x^4 + x^3 + 4}}{\sqrt[3]{x^5 - x^4 + 3}} =$$

11.

$$(3x - 2y)^2$$

## 11. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{-2x^2 + x - 4}{3x^4 + x^3 - 2} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{-4x^4 - 2x^3 + 1}{-4x^3 + x^2 + 2} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{x^4 - 2x^3 + 2}{-2x^5 + 4x^4 + 1} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{x^4 + 3x^3 + 3}{2x^3 + 3x^2 + 2} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{-2x^2 + 3x - 1}{-3x^5 + 2x^4 + 4} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{4x^4 + x^3 + 1}}{\sqrt[3]{4x^4 - x^3 - 3}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{3x^4 - 4x^3 + 3}}{\sqrt{2x^2 - 2x - 1}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{x^4 - 3x^3 - 4}}{\sqrt[4]{3x^2 + 3x + 1}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{2x^2 + x + 3}}{\sqrt[5]{2x^3 + 4x^2 + 3}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{2x^3 + 4x^2 - 3}}{\sqrt{2x^4 + x^3 + 4}} =$$

11.

$$(4x + y)^3$$

## 12. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{-3x^3 + 3x^2 - 3}{3x^5 + 2x^4 + 1} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{-4x^4 + 3x^3 + 1}{-x^5 + x^4 - 1} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{-4x^5 - 3x^4 + 2}{-2x^4 + 2x^3 + 4} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{4x^4 + x^3 + 1}{-3x^5 + x^4 + 2} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{-2x^3 - 4x^2 + 1}{3x^2 + x - 2} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{x^4 - 4x^3 + 3}}{\sqrt{x^2 + 2x - 2}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{x^3 - 3x^2 + 2}}{\sqrt[3]{x^2 - 2x - 2}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{x^4 + 2x^3 + 4}}{\sqrt{3x^2 + 2x + 4}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{2x^4 + 3x^3 + 4}}{\sqrt[4]{4x^4 + x^3 - 1}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{x^3 + x^2 - 4}}{\sqrt[4]{4x^3 + 4x^2 - 3}} =$$

11.

$$(4x + 4y)^3$$

### 13. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{3x^5 + 4x^4 - 4}{x^5 + x^4 - 2} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{-2x^2 - 3x + 1}{-x^4 + 3x^3 + 3} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{4x^4 + 3x^3 + 1}{3x^5 - 2x^4 + 3} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{-4x^2 - 4x - 4}{-3x^3 + 4x^2 + 2} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{-2x^5 + 2x^4 - 4}{2x^5 - 2x^4 - 4} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{4x^4 - 2x^3 + 1}}{\sqrt[3]{3x^3 + 4x^2 + 2}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{4x^3 + 4x^2 - 3}}{\sqrt{3x^2 + 2x + 3}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{2x^5 - 4x^4 + 2}}{\sqrt[5]{x^4 - x^3 + 2}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{3x^3 - 2x^2 - 1}}{\sqrt[3]{4x^5 - x^4 - 1}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{x^5 + x^4 - 2}}{\sqrt[4]{2x^5 + 4x^4 + 4}} =$$

11.

$$(4x + 4y)^5$$

## 14. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{4x^2 - 3x + 3}{-x^5 + 3x^4 - 4} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{-3x^5 - 3x^4 - 1}{3x^5 - 4x^4 - 3} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{4x^2 + 2x + 3}{4x^5 - 4x^4 - 4} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{4x^4 + 3x^3 - 3}{-4x^3 - 4x^2 - 4} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{2x^3 - 3x^2 - 2}{-3x^4 + x^3 - 4} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{4x^5 + 4x^4 - 3}}{\sqrt[5]{4x^2 + 3x + 4}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{4x^5 + 3x^4 - 3}}{\sqrt[5]{2x^4 - 3x^3 - 1}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{3x^2 + x - 4}}{\sqrt[3]{4x^3 + x^2 - 4}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{2x^5 - 4x^4 - 4}}{\sqrt[3]{2x^5 - 3x^4 + 1}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{3x^3 + x^2 + 4}}{\sqrt[3]{3x^3 - 4x^2 + 4}} =$$

11.

$$(3x - y)^4$$

## 15. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{4x^2 - 4x - 2}{3x^5 + 3x^4 + 2} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{-4x^4 - x^3 + 1}{-2x^4 + 4x^3 - 2} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{-2x^5 - 3x^4 + 3}{x^3 - x^2 + 4} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{-x^5 - 2x^4 + 3}{-2x^4 + 3x^3 + 2} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{3x^4 - 4x^3 - 3}{-3x^2 + x - 3} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{4x^2 - 4x - 4}}{\sqrt[5]{3x^2 + 4x - 3}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{4x^3 + 3x^2 - 1}}{\sqrt[4]{3x^5 + 4x^4 + 2}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{x^2 + 4x - 2}}{\sqrt[3]{2x^4 - 3x^3 + 4}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{x^4 - 2x^3 - 1}}{\sqrt[5]{2x^2 + x + 2}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{4x^3 + x^2 - 2}}{\sqrt[3]{4x^5 + x^4 + 3}} =$$

11.

$$(3x + 4y)^4$$

## 16. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{4x^5 + 2x^4 - 1}{-x^3 - 3x^2 - 2} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{-x^4 - x^3 + 2}{2x^3 + 4x^2 + 3} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{x^3 - x^2 + 2}{2x^3 - 4x^2 - 3} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{4x^5 + x^4 - 3}{-3x^5 + 3x^4 - 1} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{3x^5 + 2x^4 - 3}{x^5 + x^4 + 1} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{2x^4 - 3x^3 + 3}}{\sqrt[3]{2x^2 + 2x + 2}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{4x^3 - x^2 - 1}}{\sqrt[5]{x^4 + 2x^3 + 4}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{4x^4 - 3x^3 + 1}}{\sqrt[5]{2x^4 - 4x^3 - 3}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{4x^5 - 4x^4 - 4}}{\sqrt[3]{2x^3 - 4x^2 - 2}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{x^4 - 2x^3 + 3}}{\sqrt[5]{3x^4 + 3x^3 + 4}} =$$

11.

$$(3x + 3y)^5$$



## 17. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{2x^4 - x^3 - 2}{-2x^4 - 2x^3 + 2} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{3x^3 + 2x^2 - 2}{2x^5 + 3x^4 - 2} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{x^5 + 4x^4 + 4}{2x^4 - x^3 + 1} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{-x^5 - x^4 + 2}{-4x^5 + 4x^4 + 1} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{2x^2 + 4x + 1}{-2x^5 - 4x^4 + 3} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{x^3 - 4x^2 + 3}}{\sqrt[5]{3x^4 + x^3 - 4}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{4x^2 - x + 2}}{\sqrt{3x^2 + x + 3}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{4x^2 - 3x + 2}}{\sqrt[4]{4x^4 + x^3 - 4}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{3x^2 + 2x - 2}}{\sqrt{3x^3 - x^2 + 2}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{4x^2 + 4x + 4}}{\sqrt[4]{x^5 + 4x^4 - 3}} =$$

11.

$$(2x + 2y)^5$$

## 18. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{2x^5 - 2x^4 - 3}{-3x^3 - 4x^2 + 4} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{3x^2 + 3x - 4}{-2x^2 - x - 2} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{-2x^2 - 2x - 3}{-2x^3 - 2x^2 + 1} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{-3x^3 - 2x^2 - 3}{2x^5 + 4x^4 + 2} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{-4x^2 - 4x + 1}{x^3 + 2x^2 - 3} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{x^5 - 4x^4 - 2}}{\sqrt[5]{3x^3 - 3x^2 - 1}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{x^3 - x^2 - 3}}{\sqrt[4]{2x^3 + 2x^2 - 1}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{3x^5 + 2x^4 - 3}}{\sqrt{x^5 + 4x^4 + 3}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{3x^2 + x - 4}}{\sqrt[4]{2x^5 + 3x^4 - 4}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{4x^5 - x^4 + 4}}{\sqrt[3]{2x^3 + 4x^2 + 3}} =$$

11.

$$(4x + y)^4$$

## 19. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{4x^2 + 3x - 1}{4x^4 + 4x^3 - 1} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{-4x^3 + x^2 + 4}{x^2 + 2x + 3} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{-3x^3 + 4x^2 - 2}{-2x^4 + 4x^3 - 3} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{2x^2 - 2x - 1}{-4x^4 + x^3 + 4} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{-4x^4 - x^3 - 3}{-x^4 + 4x^3 - 2} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{2x^4 - 3x^3 - 1}}{\sqrt{4x^2 - 3x + 4}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{4x^3 + x^2 + 3}}{\sqrt[5]{x^4 - x^3 + 1}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{3x^4 - 3x^3 + 3}}{\sqrt[3]{3x^3 - 4x^2 - 3}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{3x^3 - 2x^2 - 1}}{\sqrt[4]{4x^5 - x^4 - 1}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{2x^2 + x + 1}}{\sqrt[5]{3x^5 - 3x^4 - 3}} =$$

11.

$$(4x - 3y)^5$$

## 20. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{-4x^3 - x^2 - 3}{-3x^2 - 4x - 4} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{-2x^3 - 3x^2 - 1}{x^3 - x^2 - 2} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{x^2 + 3x + 3}{3x^2 + 2x + 3} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{x^5 + 3x^4 - 1}{4x^3 + 4x^2 + 1} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{4x^5 - 4x^4 - 3}{-3x^3 - 3x^2 - 2} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{4x^2 + 2x - 4}}{\sqrt[5]{x^5 + x^4 - 3}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{4x^3 - x^2 + 1}}{\sqrt[4]{4x^5 - 4x^4 + 2}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{4x^2 + x - 3}}{\sqrt{4x^5 - x^4 - 3}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{x^5 - 4x^4 + 3}}{\sqrt[4]{x^2 + 4x + 1}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{4x^2 - 2x + 1}}{\sqrt[5]{3x^3 + 3x^2 - 1}} =$$

11.

$$(3x - 4y)^5$$

## 21. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{-x^2 - 2x - 1}{-3x^4 - 3x^3 - 2} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{x^3 + 3x^2 - 3}{-4x^4 - 4x^3 - 3} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{4x^4 + 4x^3 + 3}{-4x^3 - x^2 - 1} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{4x^3 + 2x^2 - 2}{3x^3 - x^2 - 1} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{-2x^2 - 4x - 4}{4x^2 + x + 1} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{4x^4 - x^3 - 3}}{\sqrt{3x^4 - 2x^3 + 3}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{x^3 + 4x^2 - 3}}{\sqrt{4x^2 - 3x + 1}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{4x^4 - 3x^3 - 2}}{\sqrt[3]{2x^5 + 2x^4 - 2}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{4x^2 - 3x - 4}}{\sqrt[4]{2x^2 - x - 1}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{3x^4 - 2x^3 - 3}}{\sqrt{4x^5 + 2x^4 - 2}} =$$

11.

$$(4x - 3y)^4$$

## 22. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{-2x^5 - x^4 + 1}{-3x^5 - x^4 + 1} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{-x^3 + 4x^2 - 3}{-3x^2 + 3x - 1} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{-x^4 - 3x^3 + 3}{4x^4 - 3x^3 + 2} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{x^3 - 4x^2 + 1}{3x^5 + 3x^4 - 1} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{2x^4 + x^3 + 2}{2x^3 + 4x^2 + 2} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{3x^5 - x^4 + 3}}{\sqrt[3]{4x^3 + 4x^2 - 2}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{x^3 + 3x^2 + 3}}{\sqrt{4x^2 + 4x + 3}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{2x^5 + 2x^4 + 2}}{\sqrt{3x^2 + 2x - 1}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{2x^5 - 4x^4 + 4}}{\sqrt[3]{x^3 + 4x^2 + 4}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{x^2 + 3x - 2}}{\sqrt[3]{x^3 - x^2 + 2}} =$$

11.

$$(2x + y)^5$$

## 23. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{-2x^2 + 4x - 3}{-4x^4 - 3x^3 + 1} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{-2x^2 - 4x - 2}{3x^4 - x^3 - 3} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{2x^2 + x + 3}{4x^3 + x^2 + 4} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{-x^5 - 4x^4 - 1}{2x^5 + 3x^4 + 4} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{-3x^2 - 4x + 1}{-x^4 - 3x^3 - 2} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{2x^5 - 3x^4 - 2}}{\sqrt{2x^4 + 3x^3 + 2}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{4x^3 + 2x^2 - 3}}{\sqrt[5]{4x^3 - 4x^2 - 2}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{4x^5 - 3x^4 - 3}}{\sqrt{3x^5 + 4x^4 - 2}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{2x^5 + 2x^4 + 1}}{\sqrt[4]{3x^2 + x - 4}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{2x^5 - 4x^4 - 1}}{\sqrt[5]{3x^4 - 3x^3 + 2}} =$$

11.

$$(2x - 3y)^5$$

## 24. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{-2x^2 + 2x + 4}{2x^3 + 3x^2 + 3} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{3x^4 + 4x^3 + 3}{x^2 + 4x + 4} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{4x^2 - 4x - 1}{3x^5 - 3x^4 - 1} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{2x^4 + 3x^3 - 4}{3x^5 + 4x^4 + 1} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{x^2 + 4x - 4}{-3x^4 - 4x^3 + 3} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{x^4 - 3x^3 - 2}}{\sqrt[3]{4x^5 - 4x^4 + 2}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{4x^2 + 4x - 4}}{\sqrt{4x^5 + 3x^4 - 1}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{2x^5 - 3x^4 - 4}}{\sqrt[5]{3x^5 + x^4 + 1}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{3x^2 - x + 3}}{\sqrt{4x^5 - 4x^4 + 2}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{4x^4 - 3x^3 + 2}}{\sqrt[4]{4x^5 + 2x^4 - 4}} =$$

11.

$$(2x - y)^4$$



## 25. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{-3x^4 - x^3 + 1}{-x^2 + 2x - 1} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{-3x^2 + 3x + 4}{2x^5 + 4x^4 - 2} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{-x^3 - 2x^2 - 3}{3x^2 + 4x - 4} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{-x^5 - x^4 - 4}{-2x^4 + 2x^3 - 2} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{-4x^5 - 3x^4 - 4}{2x^2 - 3x - 1} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{4x^2 + 2x - 3}}{\sqrt[4]{2x^5 + x^4 + 4}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{4x^4 - 3x^3 - 3}}{\sqrt[3]{4x^5 + 3x^4 - 1}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{3x^2 + 4x - 2}}{\sqrt{x^3 + 2x^2 - 3}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{4x^3 - 2x^2 - 2}}{\sqrt[5]{3x^2 - x + 3}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{x^4 + x^3 + 4}}{\sqrt{3x^3 - 2x^2 - 3}} =$$

11.

$$(3x + 2y)^4$$

## 26. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{4x^3 + x^2 - 3}{4x^2 - 3x - 2} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{x^2 + 4x + 3}{x^4 + 3x^3 + 3} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{4x^2 + 3x + 4}{x^3 + 4x^2 + 1} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{-4x^2 + 3x + 4}{x^5 - 4x^4 - 3} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{x^3 + 2x^2 - 2}{2x^3 - x^2 - 4} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{3x^4 - 2x^3 - 1}}{\sqrt[4]{3x^3 - 2x^2 - 1}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{3x^3 + x^2 - 2}}{\sqrt{4x^4 + 4x^3 - 2}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{4x^2 - 4x - 1}}{\sqrt{x^5 - 2x^4 - 4}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{3x^5 + 4x^4 - 1}}{\sqrt[3]{3x^4 + 2x^3 - 2}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{4x^5 + 4x^4 - 1}}{\sqrt[4]{4x^5 + x^4 + 1}} =$$

11.

$$(4x - 2y)^3$$

## 27. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{x^5 - 3x^4 + 1}{-2x^2 + 3x - 3} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{x^2 - 2x + 4}{-4x^5 - 2x^4 - 3} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{-3x^3 - x^2 - 2}{-4x^5 - 4x^4 + 1} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{3x^5 - 4x^4 + 1}{2x^4 - x^3 + 2} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{x^4 + 3x^3 + 1}{-x^2 - 4x + 3} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{x^4 - 2x^3 + 2}}{\sqrt{2x^3 - x^2 - 4}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{2x^2 + x - 1}}{\sqrt[3]{x^3 + 3x^2 - 1}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{3x^4 + 4x^3 - 3}}{\sqrt{2x^2 + 2x + 2}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{2x^3 - 3x^2 - 1}}{\sqrt[3]{x^2 + 2x + 1}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{2x^3 + 2x^2 + 2}}{\sqrt[5]{x^3 + 3x^2 + 4}} =$$

11.

$$(x + 2y)^4$$

## 28. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{-2x^2 - 4x - 3}{-3x^2 + 2x + 2} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{-2x^2 - 2x + 2}{2x^2 + 2x - 4} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{-2x^3 - 3x^2 - 2}{x^4 - 3x^3 - 3} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{-3x^5 - 3x^4 + 3}{x^3 + 4x^2 - 4} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{3x^5 + x^4 + 1}{-4x^2 - 3x + 3} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{4x^3 - 3x^2 + 4}}{\sqrt{x^4 - x^3 + 1}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{4x^4 - x^3 - 1}}{\sqrt{3x^4 - 3x^3 - 4}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{2x^5 + 4x^4 - 3}}{\sqrt[4]{x^4 + 3x^3 - 1}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{4x^3 + 3x^2 - 1}}{\sqrt[4]{3x^2 - 4x + 2}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{4x^3 + 2x^2 - 3}}{\sqrt[5]{4x^3 + 4x^2 - 3}} =$$

11.

$$(3x - 2y)^3$$

## 29. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{x^5 - 4x^4 + 2}{-2x^3 + 4x^2 - 2} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{x^5 - 4x^4 - 4}{-4x^5 + 2x^4 - 3} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{-4x^2 + 3x - 2}{4x^2 - 2x - 3} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{-x^3 - x^2 + 3}{4x^4 - x^3 + 4} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{-2x^3 - 2x^2 + 2}{2x^4 + x^3 + 2} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{3x^4 - 4x^3 - 2}}{\sqrt[3]{2x^4 - 4x^3 - 4}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{3x^2 - 4x - 2}}{\sqrt{3x^2 + x + 3}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{4x^2 + 3x - 2}}{\sqrt[3]{2x^5 + 3x^4 + 4}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{x^3 + 2x^2 + 3}}{\sqrt[4]{3x^3 + 2x^2 - 4}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{4x^5 - 4x^4 - 2}}{\sqrt[4]{x^5 - x^4 - 4}} =$$

11.

$$(2x - 2y)^3$$

## 30. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{3x^3 + x^2 + 1}{3x^4 + x^3 - 3} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{4x^2 - x + 2}{-3x^4 + 4x^3 + 3} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{2x^3 - 2x^2 - 1}{3x^3 + x^2 - 3} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{-3x^3 + 2x^2 - 2}{3x^4 + 4x^3 + 4} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{2x^5 + 3x^4 - 4}{x^3 - x^2 + 2} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{2x^3 - 4x^2 - 3}}{\sqrt{x^2 + 2x + 4}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{2x^2 - 4x - 1}}{\sqrt[4]{2x^3 + 2x^2 + 3}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{3x^2 - x - 3}}{\sqrt{x^5 + 4x^4 - 1}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{4x^5 - x^4 + 1}}{\sqrt[5]{4x^3 - 2x^2 - 3}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{3x^5 + 4x^4 + 1}}{\sqrt[4]{x^4 - 4x^3 + 4}} =$$

11.

$$(4x + y)^4$$

## 31. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{-4x^5 - 4x^4 - 4}{4x^5 - 4x^4 + 4} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{3x^2 + 2x + 3}{-4x^2 - 4x - 3} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{-x^4 + x^3 + 2}{x^4 - x^3 - 3} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{4x^5 - 4x^4 - 4}{2x^5 + x^4 + 3} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{-4x^5 + x^4 - 1}{-4x^3 + 3x^2 + 4} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{3x^4 + 3x^3 - 2}}{\sqrt[3]{x^5 - x^4 - 2}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{x^3 + 2x^2 + 2}}{\sqrt[3]{4x^5 - 2x^4 - 4}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{3x^5 + 2x^4 + 2}}{\sqrt[5]{4x^2 - 3x - 1}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{3x^4 + 2x^3 + 1}}{\sqrt[3]{4x^5 - 2x^4 - 2}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{3x^2 - x - 2}}{\sqrt[5]{2x^2 + 4x - 1}} =$$

11.

$$(x - 2y)^5$$

## 32. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{-x^3 - x^2 - 2}{-x^5 + 4x^4 + 3} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{-x^4 + 2x^3 - 4}{4x^4 + 2x^3 - 3} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{x^5 + x^4 - 4}{4x^5 - 2x^4 + 1} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{-2x^3 - 4x^2 - 2}{x^5 - 3x^4 - 4} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{-2x^3 + 4x^2 - 4}{x^4 + 4x^3 - 2} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{4x^5 - x^4 + 1}}{\sqrt[4]{4x^4 + 3x^3 - 4}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{3x^5 + 2x^4 - 2}}{\sqrt[4]{4x^4 + 3x^3 - 4}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{2x^2 - 3x + 2}}{\sqrt[4]{3x^5 + 2x^4 - 1}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{4x^4 - 3x^3 - 1}}{\sqrt[5]{2x^4 - 3x^3 - 1}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{4x^5 + x^4 + 4}}{\sqrt[4]{x^2 + 4x - 4}} =$$

11.

$$(2x + 3y)^5$$



### 33. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{x^4 + x^3 - 1}{-4x^4 - 3x^3 + 3} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{-2x^3 - 3x^2 + 4}{x^2 + 3x - 2} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{x^4 + x^3 + 4}{3x^5 + 4x^4 + 1} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{-4x^2 + 4x - 2}{-2x^5 + x^4 + 1} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{2x^5 - 4x^4 - 1}{-x^2 + x + 1} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{2x^2 + x - 3}}{\sqrt{4x^2 + x - 1}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{4x^2 - x - 4}}{\sqrt{2x^5 + 2x^4 + 1}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{2x^2 + 4x - 3}}{\sqrt{3x^3 - 2x^2 + 1}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{3x^2 + 3x - 1}}{\sqrt[3]{2x^2 - x + 4}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{2x^5 - 3x^4 + 3}}{\sqrt[3]{4x^4 + 2x^3 - 3}} =$$

11.

$$(4x - 2y)^2$$

## 34. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{x^4 + 2x^3 - 1}{4x^4 - 4x^3 + 3} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{4x^4 + 4x^3 + 1}{4x^2 - x + 1} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{3x^5 - x^4 + 3}{2x^4 + x^3 + 2} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{2x^2 + 3x - 3}{x^4 + 3x^3 - 1} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{-x^2 + x - 2}{-3x^4 + 4x^3 + 1} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{x^2 + x + 4}}{\sqrt{x^2 - x + 2}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{4x^2 + x - 2}}{\sqrt[5]{x^2 + 4x - 4}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{x^5 - 4x^4 - 2}}{\sqrt[4]{4x^4 - 3x^3 - 1}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{2x^3 - 3x^2 - 2}}{\sqrt[4]{x^3 - x^2 + 3}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{x^4 + 2x^3 - 2}}{\sqrt[4]{3x^3 - 4x^2 - 1}} =$$

11.

$$(2x + 2y)^3$$

## 35. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{-4x^5 - 3x^4 + 3}{4x^2 + 4x + 1} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{-3x^5 - 3x^4 + 4}{-2x^3 + 4x^2 + 4} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{-2x^2 - 4x - 4}{2x^2 + x - 1} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{-x^2 - x - 3}{-3x^5 - 3x^4 - 4} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{-3x^4 - x^3 + 3}{-x^2 - 4x + 4} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{2x^3 - x^2 + 3}}{\sqrt{2x^4 - 3x^3 - 3}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{2x^2 - 3x - 2}}{\sqrt[4]{3x^3 + 4x^2 - 3}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{2x^2 - x + 3}}{\sqrt{3x^3 + x^2 - 3}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{2x^3 + x^2 - 1}}{\sqrt[5]{2x^4 + 4x^3 - 2}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{x^5 + 3x^4 + 2}}{\sqrt[5]{x^5 + 2x^4 - 2}} =$$

11.

$$(2x - 3y)^2$$

## 36. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{2x^4 + 4x^3 + 3}{-4x^3 + x^2 + 3} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{x^3 - 3x^2 + 4}{x^3 - 3x^2 + 4} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{4x^2 + 2x + 2}{-3x^2 - x - 3} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{-x^3 + 3x^2 + 4}{x^4 + 2x^3 + 2} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{2x^4 + 3x^3 + 2}{x^3 + 2x^2 + 2} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{4x^3 + 2x^2 + 3}}{\sqrt[5]{3x^4 - x^3 - 1}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{4x^3 + 2x^2 + 2}}{\sqrt[3]{4x^4 - 3x^3 + 2}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{2x^5 + 3x^4 - 4}}{\sqrt[5]{3x^5 + 3x^4 + 1}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{x^4 - 2x^3 - 1}}{\sqrt[5]{3x^3 - 2x^2 + 2}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{x^2 - 3x + 3}}{\sqrt[5]{4x^4 + 4x^3 - 3}} =$$

11.

$$(2x - 2y)^3$$

## 37. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{-3x^4 + x^3 - 4}{2x^5 + x^4 - 2} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{3x^2 + x + 2}{-4x^5 + 2x^4 + 2} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{3x^4 + x^3 + 4}{-3x^3 + x^2 - 1} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{3x^3 + 4x^2 - 4}{3x^5 + 3x^4 - 4} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{2x^5 - x^4 - 2}{4x^4 + x^3 - 3} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{4x^4 - 3x^3 - 1}}{\sqrt[3]{3x^5 + 2x^4 - 4}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{4x^2 + 3x + 4}}{\sqrt[5]{4x^2 + x + 2}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{x^3 + 2x^2 - 4}}{\sqrt[3]{x^3 + 3x^2 - 1}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{3x^3 + 4x^2 - 4}}{\sqrt[5]{x^2 - 2x + 1}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{3x^5 + x^4 - 1}}{\sqrt[3]{4x^3 + x^2 - 4}} =$$

11.

$$(2x + 4y)^2$$

## 38. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{-3x^2 - 4x - 4}{4x^3 - 4x^2 - 2} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{x^4 - 2x^3 - 2}{3x^5 - 2x^4 - 1} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{-x^5 - 4x^4 - 3}{-2x^2 - x + 3} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{-x^4 - 4x^3 - 1}{-2x^5 + 4x^4 - 4} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{3x^5 - 2x^4 + 1}{4x^2 - x + 4} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{2x^5 - 3x^4 + 3}}{\sqrt[5]{2x^3 - x^2 + 3}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{2x^5 + 3x^4 + 4}}{\sqrt[4]{x^5 - 4x^4 + 4}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{3x^5 + x^4 + 3}}{\sqrt[5]{3x^5 - 3x^4 + 2}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{4x^2 - x + 4}}{\sqrt[5]{3x^3 + 4x^2 + 4}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{3x^5 - 3x^4 + 2}}{\sqrt[4]{4x^2 + x - 1}} =$$

11.

$$(x + y)^5$$

## 39. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{-3x^3 - 3x^2 - 1}{-3x^4 + x^3 + 2} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{-2x^4 + x^3 + 1}{-4x^4 - 4x^3 + 3} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{-x^2 - 3x - 4}{3x^2 + 3x - 4} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{x^2 - 3x - 4}{x^2 + 2x - 3} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{-4x^4 - 3x^3 + 4}{-2x^5 - 2x^4 - 3} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{x^4 - 4x^3 + 2}}{\sqrt[3]{3x^5 - 3x^4 + 4}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{2x^3 - x^2 + 2}}{\sqrt[3]{x^4 + 3x^3 - 2}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{x^5 + 4x^4 + 3}}{\sqrt{x^4 - x^3 - 1}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{x^5 - 4x^4 + 4}}{\sqrt[3]{3x^2 + 2x + 2}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{2x^2 - 3x + 2}}{\sqrt[4]{4x^4 - x^3 - 4}} =$$

11.

$$(4x + y)^3$$

## 40. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{x^4 - 2x^3 + 2}{-2x^4 + 2x^3 - 1} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{-4x^4 + x^3 - 4}{3x^2 - 3x + 1} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{4x^3 - 2x^2 + 1}{-3x^3 + x^2 + 4} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{4x^4 - 2x^3 + 3}{x^2 - 2x - 2} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{-3x^2 - x - 1}{-2x^4 - x^3 + 3} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{2x^2 - 2x + 1}}{\sqrt[3]{3x^3 + 4x^2 - 2}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{3x^2 - 2x - 1}}{\sqrt[5]{x^4 + 4x^3 + 2}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{3x^4 + 3x^3 + 1}}{\sqrt[5]{3x^2 + 3x - 4}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{4x^3 + 2x^2 + 3}}{\sqrt[5]{3x^2 - 3x + 1}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{2x^3 + 2x^2 + 4}}{\sqrt[4]{3x^2 - 3x - 4}} =$$

11.

$$(x + 3y)^3$$



## 41. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{2x^2 + 3x + 4}{x^3 + 4x^2 - 2} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{4x^4 - 3x^3 + 2}{2x^5 - 3x^4 - 3} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{-x^5 - x^4 - 1}{-3x^2 - 4x + 3} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{2x^3 - x^2 + 4}{-3x^5 + 2x^4 - 4} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{3x^3 - 2x^2 - 3}{x^3 + x^2 + 1} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{x^4 - 3x^3 - 3}}{\sqrt[5]{4x^5 + x^4 - 2}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{x^3 - 3x^2 - 2}}{\sqrt[4]{4x^2 - 3x + 1}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{3x^2 - 4x - 4}}{\sqrt{x^3 + 2x^2 + 3}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{3x^5 - 2x^4 + 2}}{\sqrt[3]{3x^2 + 3x - 3}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{2x^5 - 3x^4 - 1}}{\sqrt[4]{4x^5 + 2x^4 + 3}} =$$

11.

$$(3x - y)^5$$

## 42. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{3x^4 + x^3 - 3}{2x^2 + 3x + 2} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{-x^3 + 3x^2 - 2}{-2x^3 - 3x^2 + 4} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{-3x^3 - 3x^2 - 3}{3x^2 - 4x - 2} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{-x^5 - 2x^4 + 3}{2x^3 - x^2 - 2} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{3x^2 - 3x - 1}{-x^5 - 2x^4 + 2} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{3x^2 - x + 1}}{\sqrt[5]{4x^3 + 4x^2 + 4}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{3x^2 + 3x + 1}}{\sqrt[5]{x^3 + 2x^2 - 4}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{2x^3 - 2x^2 + 2}}{\sqrt[4]{x^2 - x - 2}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{4x^4 + 3x^3 + 1}}{\sqrt[4]{4x^3 + x^2 - 4}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{3x^5 - 4x^4 + 2}}{\sqrt[3]{x^2 + x - 2}} =$$

11.

$$(4x + y)^2$$

## 43. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{3x^2 + 3x + 2}{4x^5 - x^4 - 1} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{-x^3 + 4x^2 + 4}{4x^3 + 3x^2 - 2} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{-3x^5 - 4x^4 - 2}{-3x^5 + 4x^4 + 2} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{x^2 - 3x + 3}{4x^5 - 4x^4 + 1} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{2x^2 + 3x - 1}{x^4 - x^3 - 2} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{2x^3 - 3x^2 - 3}}{\sqrt[4]{2x^5 + x^4 - 4}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{3x^3 - x^2 + 4}}{\sqrt[4]{4x^3 - x^2 - 4}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{4x^2 + 3x + 1}}{\sqrt[5]{4x^2 - 2x - 2}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{3x^4 - 3x^3 + 3}}{\sqrt[5]{4x^2 + 2x - 3}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{2x^3 + x^2 - 3}}{\sqrt[5]{4x^2 - x + 3}} =$$

11.

$$(4x + 4y)^4$$

## 44. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{-3x^4 + 2x^3 + 3}{2x^5 - 3x^4 + 4} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{-4x^3 + 2x^2 + 3}{2x^5 - 4x^4 + 1} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{-2x^2 + x - 1}{3x^3 + 2x^2 + 4} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{-2x^2 + x + 2}{2x^5 - x^4 + 2} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{-3x^2 - x - 2}{x^2 + 2x + 4} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{2x^3 - 4x^2 - 2}}{\sqrt[4]{2x^3 - 2x^2 - 4}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{2x^3 - x^2 + 4}}{\sqrt[5]{2x^5 - 4x^4 - 4}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{2x^3 + 4x^2 - 2}}{\sqrt[5]{x^2 + x + 2}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{2x^5 + x^4 - 3}}{\sqrt[3]{4x^2 - x + 3}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[5]{x^5 - 3x^4 + 1}}{\sqrt[4]{x^5 + x^4 + 2}} =$$

11.

$$(2x - 4y)^2$$

## 45. Ejercicios

1.

$$\lim_{x \rightarrow +\infty} \frac{4x^5 + 4x^4 - 2}{-x^3 + x^2 + 2} =$$

2.

$$\lim_{x \rightarrow +\infty} \frac{-3x^4 - 2x^3 - 3}{4x^4 + 2x^3 + 1} =$$

3.

$$\lim_{x \rightarrow +\infty} \frac{-x^5 + 3x^4 - 1}{4x^4 - x^3 - 4} =$$

4.

$$\lim_{x \rightarrow +\infty} \frac{2x^2 - 2x - 4}{4x^3 + 2x^2 + 1} =$$

5.

$$\lim_{x \rightarrow +\infty} \frac{-4x^4 - 3x^3 - 1}{3x^2 + 4x - 3} =$$

6.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{x^4 + 4x^3 - 4}}{\sqrt[5]{2x^4 + 2x^3 + 4}} =$$

7.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{3x^4 - 4x^3 - 2}}{\sqrt[5]{3x^3 + x^2 + 2}} =$$

8.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{4x^4 - x^3 + 4}}{\sqrt[5]{x^5 - x^4 - 3}} =$$

9.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{4x^4 + 4x^3 - 4}}{\sqrt[5]{4x^4 + 3x^3 + 2}} =$$

10.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt[3]{x^3 + 2x^2 + 1}}{\sqrt[4]{x^3 - 2x^2 + 2}} =$$

11.

$$(3x + 3y)^5$$